

CSCN8020 Assignment 3

Deep Q-Network Control of the Unitree G1 Left Elbow

Student information

FULL NAME Viraj Dipakkumar Mistry
STUDENT ID 9088985
COURSE CSCN8020 - Reinforcement Learning

Project summary

This project implements a student-written PyTorch Deep Q-Network to control the Unitree G1 left elbow in MuJoCo. The agent maps four observation values to three discrete actions and learns with replay memory, epsilon-greedy exploration, Bellman updates, and a target network. Two controlled exploration-decay experiments were trained headlessly on CPU and evaluated at four benchmark angles. Both achieved 20/20 greedy successes. Configuration A was selected for its higher mean reward and lower final error. The repository contains reproducible source code, checkpoints, structured metrics, plots, tests, a completed notebook, technical report, and a saved-policy video demonstrating multiple goals without retraining.

Repository links

GITHUB REPOSITORY

https://github.com/vir33mistry/CSCN8020_Assignment3

CLONEABLE URL

https://github.com/vir33mistry/CSCN8020_Assignment3.git

Selected result: 20/20 greedy successes (100%)

Configuration A | Mean reward 13.1497 | Mean final error 0.0071 rad